

ELIZABETH WRABACK
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EDUCATION

University of Michigan, Ann Arbor, MI, USA

Ph.D. in Climate and Space Sciences and Engineering

Sep. 2025

Thesis Title: *Non-Equilibrium Ionization and Spectroscopic Modeling of Coronal Mass Ejections*

Advisors: *Enrico Landi, Ward (Chip) Manchester*

Master of Science: Climate and Space Sciences and Engineering

Aug. 2023

University of Maryland, College Park, MD, USA

Bachelor of Science: Physics and Astronomy

May 2021

Graduated with High Honors in Physics and with High Honors in Astronomy

Honors Thesis Title: *Microphysics of Localized Structures in the Magnetosphere from Multispacecraft MMS Data*

Advisors: *Surjalal Sharma, Jason Shuster*

RESEARCH INTERESTS

My research centers on the how we can connect the low solar corona to the near-Earth environment with a focus on eruptive events. I integrate ground-based and space-based observations with magnetohydrodynamics modeling to connect the different observations and to understand how, where, and when energy is in the corona and through the heliosphere.

PROFESSIONAL EXPERIENCE

High Altitude Observatory, NSF National Center for Atmospheric Research

Boulder, CO, USA

Advanced Study Program Postdoctoral Fellow

Oct. 2025-present

Hosts: *Sarah Gibson, Yuhong Fan*

Solar Physics Laboratory, NASA Goddard Space Flight Center

Greenbelt, MD, USA

Undergraduate Intern

Summers 2019, 2020, 2021

Host: *C. Nick Arge*

PUBLICATIONS

Wraback, E.M., Manchester, W.B., Landi, E., & Szente, J. 3D Non-Equilibrium Ionization and Spectroscopic Modeling of Coronal Mass Ejections III - Charge State Evolution. (2026) *ApJ*. 996, 65. <https://doi.org/10.3847/1538-4357/ae1a7a>

Wraback, E.M., Manchester, W.B., Landi, E., & Szente, J. 3D Non-Equilibrium Ionization and Spectroscopic Modeling of Coronal Mass Ejections II - CME Energy Budget. (2025) *ApJ*. 995, 105. <https://doi.org/10.3847/1538-4357/ae18a1>

Wraback, E.M., Landi, E., Manchester, W.B., & Szente, J. 3D Non-Equilibrium Ionization and Spectroscopic Modeling of Coronal Mass Ejections I - Comparison with *Hinode*/EIS Observations. (2025). *ApJ*. 980, 30. <https://doi.org/10.3847/1538-4357/ada7e8>

Wraback, E.M., Landi, E., & Manchester, W.B. EIS Spectral Atlas of Cool Prominence Material in CME Core. (2024). *ApJ*. 970, 182. <https://doi.org/10.3847/1538-4357/ad625f>

Wraback, E.M., Landi, E., & Manchester, W.B. Using the Cartwheel CME to Predict Off-Limb Observations of CMEs for New and Upcoming UV and EUV Spectrometers. (2024). *ApJ*. 974, 32. <https://doi.org/10.3847/1538-4357/ad6d58>

Wraback, E.M., Hoffmann, A.P., Manchester, W.B., Sokolov, I.V., van der Holst, B., & Carpenter, D. Simulating Compressive Stream Interaction Regions during Parker Solar Probe’s First Perihelion using the Stream-Aligned Magnetohydrodynamics. (2024). *ApJ*. 962, 2. <https://doi.org/10.3847/1538-4357/ad21fd>

In-preparation

Wraback, E.M., Farid, S., Schad, T., Gibson, S.E., Fan, Y., de Toma, G., et al. Measurement of Stokes V in a Coronal Cavity using DKIST/CryoNIRSP. *In-prep for submission to ApJL*.

FELLOWSHIPS & GRANTS

NSF NCAR Advanced Study Program Postdoctoral Fellowship

Oct. 2025 - Oct. 2027

TEACHING

Preparing Future Faculty Program Certificate	June 2025
University of Michigan Center for Research on Learning and Teaching	
Grader Space Plasma Physics (SPACE 597)	Fall 2024
Grader Introduction to Space Physics (SPACE 574)	Fall 2023

SERVICE

Reviewer for ApJ, RSTA	
SHINE Student Representative	2023-2025
Michigan Geophysical Union Organizing Committee, Co-Chair	2023-2024
Michigan Geophysical Union Organizing Committee, Judging Subcommittee Chair	2021-2023
Peer Mentor (as undergraduate and graduate student)	2019-Present

AWARDS

Dr. Richard R. Babcock, Jr. Graduate Student Award (departmental award)	Apr. 2024
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INVITED PRESENTATIONS

<i>Understanding the Evolution of CME Plasmas in the Solar Corona from EUV Spectroscopy and Charge State Composition</i>	
Max-Planck Institute for Solar System Research	Dec. 2024
Rice University Space Physics Seminar	Nov. 2024
High-Altitude Observatory Colloquium Series	May 2024
<i>Simulating Compressive Stream Interaction Regions during Parker Solar Probe’s First Perihelion Using Stream-aligned Magnetohydrodynamics</i>	
Solar Orbiter Origins and Connection Science Working Group	Sep. 2024
<i>Simulating the Cartwheel CME’s Early Evolution with Non-Equilibrium Charge States for Comparison to High-Resolution EUV Spectroscopic Observations</i>	
Solar Connectivity ISFM Team Meeting at NASA GSFC	Feb. 2024
PSP Large-Scale Structure Working Group	Jan. 2024

ORAL PRESENTATIONS

<i>Simulating the Cartwheel CME’s Early Evolution with Non-Equilibrium Charge States for Comparison to High-Resolution EUV Spectroscopic Observations</i>	
EGU General Assembly 2024	Apr. 2024
Triennial Earth-Sun Summit (TESS) 2024	Apr. 2024
2023 AGU Fall Meeting	Dec. 2023
<i>Using the Cartwheel CME to Predict EUV Off-Limb Observations of CME Cores for SPICE and EUVST</i>	
2023 AGU Fall Meeting	Dec. 2023

U.S. Solar Orbiter Science Team Meeting	Oct. 2023
<i>Simulating Stream Interaction Regions during Parker Solar Probe's First Approach with the Alfvén Wave Solar atmosphere Model</i>	
The Fifth Parker Solar Probe Scholars Meeting	May 2023
<i>Utilizing Electron Heat Flux Data to Improve IMF Polarity Specification of the Wang-Sheeley-Arge Model</i>	
2021 AGU Fall Meeting	Dec. 2021

POSTER PRESENTATIONS

<i>Charge States: An Insight into the CME Energy Budget</i>	
2025 SHINE Meeting	June 2025
<i>Simulations and Diagnostics of CME Charge State Evolution from Transition Region through Heliosphere</i>	
2024 AGU Fall Meeting	Dec. 2024
2024 SHINE Meeting	Aug. 2024
<i>Michigan Geophysical Union: Creating a Space to Empower Students in Earth and Space Sciences</i>	
2024 AGU Fall Meeting	Dec. 2024
<i>Simulating the Cartwheel CME for Comparison to High-Resolution EUV Spectroscopy</i>	
2023 SHINE Meeting	Aug. 2023
<i>High-Resolution EUV Spectroscopy of CME Onset and Early Evolution</i>	
2022 AGU Fall Meeting	Dec. 2022
<i>Simulating Stream Interaction Regions during Parker Solar Probe's First Approach with the Alfvén Wave Solar atmosphere Model</i>	
2022 SHINE Meeting	Jun. 2022
<i>Utilizing Electron Heat Flux Data to Improve IMF Polarity Specification of the Wang-Sheeley-Arge Model</i>	
2021 SHINE Meeting	Aug. 2021
<i>Microphysics of Localized Structures in the Magnetosphere from Multispacecraft MMS Data</i>	
2021 AGU Fall Meeting	Dec. 2021

MEMBERSHIPS

- American Geophysical Union (AGU)
- AAS/Solar Physics Division (AAS/SPD)

Last Updated Jan. 2026